

The Equation of a Hyperplane

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Where this fits: Pipeline map step 0 (Foundations). See the [Course map](#).



- **Builds on:** Dot product ([Note 362](#)).
- **Leads to:** Perceptron ([Note 1004](#)); Perceptron trick ([Note 1005](#)).

1. Overview

Key point: One equation, $w^T x + w_0 = 0$, describes a line in 2D, a plane in 3D and a hyperplane in any number of dimensions; w is perpendicular to it and w_0 moves it away from the origin.

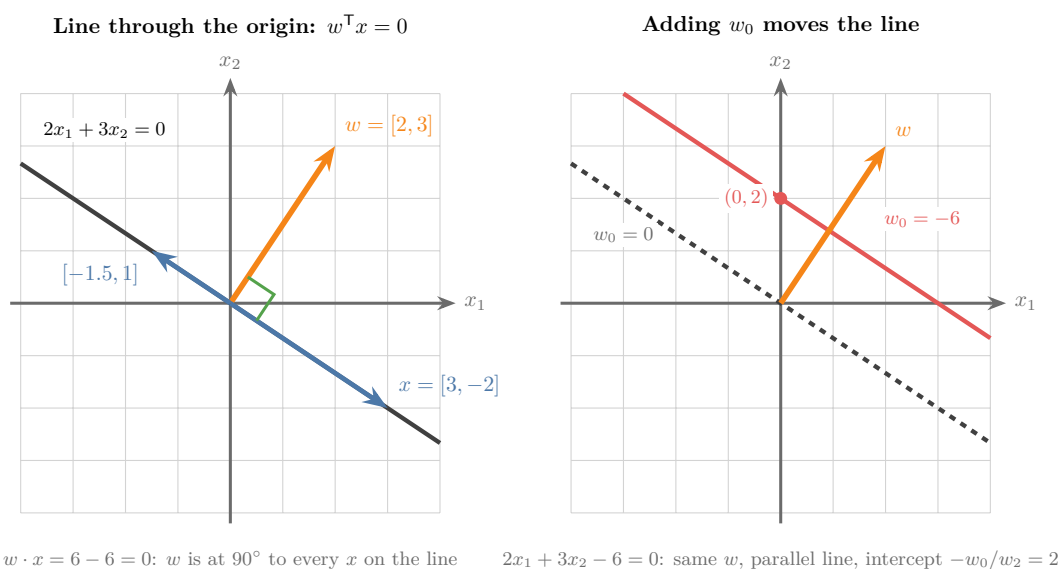


Figure 1: Left: a line through the origin and the vector w perpendicular to it. Right: adding w_0 moves the line without turning it

Figure 1 shows the whole result in 2D. On the left, every vector x on the line makes a right angle with w , so $w \cdot x = 0$. On the right, adding a number w_0 slides the line to a parallel position. This Note builds that equation step by step from the school equation of a line, and then reads off what w and w_0 mean.

It uses the dot product and its geometric form from the [dot product and cosine similarity Note](#).

2. Why ML needs hyperplanes

Key point: A line in 2D becomes a plane in 3D and a hyperplane beyond; models that classify with a straight boundary need its equation in any dimension.

In 2D a flat boundary is a line; in 3D it is a plane. In 4D, 5D or n D the equivalent is a hyperplane (see the [multiple linear regression Note](#)).

ML data is very often high-dimensional. Even the iris toy dataset has 4 input columns, and MNIST, the handwritten digits dataset, has 784 (one per pixel of a 28 x 28 image). Classifiers that separate classes with a straight boundary, such as [logistic regression](#) and [SVM](#), therefore work with hyperplanes. To use them, and to code such algorithms ourselves, we need one equation that works in every dimension.

3. From a line to a hyperplane

Key point: Rename the general form of a line as $w_1x_1 + w_2x_2 + w_0 = 0$; each extra dimension adds one more term w_ix_i .

3.1 The general form of a line

Key point: $y = mx + b$ and $ax + by + c = 0$ describe the same line; the general form is the one that grows into higher dimensions.

In school a line is $y = mx + b$, with slope m and intercept b . The same line has a general form, $ax + by + c = 0$, used in the [perceptron trick Note](#) (section 3.1).

Solving the general form for y links the two:

$$y = -\frac{a}{b}x - \frac{c}{b}, \quad \text{so} \quad m = -\frac{a}{b}, \quad \text{intercept} = -\frac{c}{b}$$

(the b of the general form is a different number from the intercept b of $y = mx + b$). We use the general form, because it treats every axis the same way.

3.2 Renaming for n dimensions

Key point: Call the axes $x_1, x_2, \dots$ and the coefficients $w_1, w_2, \dots$, with w_0 for the constant.

Letters x, y, z run out after three dimensions, so we number the axes instead: $x_1, x_2, x_3, \dots$. We number the coefficients the same way, $w_1, w_2, \dots$, and call the constant term w_0 . Then:

- **Line (2D):** $w_1x_1 + w_2x_2 + w_0 = 0$
- **Plane (3D):** $w_1x_1 + w_2x_2 + w_3x_3 + w_0 = 0$
- **Hyperplane (n D):** $w_1x_1 + w_2x_2 + \dots + w_nx_n + w_0 = 0$

Each new dimension adds one term. With numbers, the line $2x + 3y - 6 = 0$ becomes $2x_1 + 3x_2 - 6 = 0$: here $w_1 = 2$, $w_2 = 3$ and $w_0 = -6$.

4. The vector form

Key point: The sum $w_1x_1 + \dots + w_nx_n$ is a dot product, so the hyperplane is $w \cdot x + w_0 = 0$, or $w^\top x + w_0 = 0$.

Look at the part $w_1x_1 + w_2x_2 + \dots + w_nx_n$. It multiplies matching components and adds them: it is a dot product. Collect the coefficients into one vector and the coordinates into another:

$$w = \begin{bmatrix} w_1 \\ w_2 \\ \vdots \\ w_n \end{bmatrix}, \quad x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

Both are column vectors, the default. Writing the dot product as a row times a column, $w \cdot x = w^\top x$, gives the equation of a hyperplane:

1. **In words:** the dot product of the coefficient vector with any point on the hyperplane, plus the constant, is zero.
2. **Formula:**

$$w^\top x + w_0 = 0$$

3. **Example:** for $w = [2, 3]$ and $w_0 = -6$, the point $x = [3, 0]$ gives

$$w^\top x + w_0 = 2 \times 3 + 3 \times 0 - 6 = 0$$

so $[3, 0]$ lies on the line $2x_1 + 3x_2 - 6 = 0$.

This one equation is valid in 2D, 3D and n D: only the number of components of w and x changes. It is the same $w \cdot u + b$ that SVM uses in the [SVM maths Note](#), with $b = w_0$.

5. What w_0 means

Key point: w_0 shifts the hyperplane away from the origin; $w_0 = 0$ exactly when the hyperplane passes through the origin.

To see what w_0 does, go back to 2D, where the line is both $x_2 = mx_1 + b$ and $w_1x_1 + w_2x_2 + w_0 = 0$. Section 3.1 matched the intercepts:

$$b = -\frac{w_0}{w_2}$$

Now slide the line down until it passes through the origin. Its intercept b becomes 0, and since $w_2 \neq 0$, that forces $w_0 = 0$. So:

- A line, plane or hyperplane passes through the origin exactly when $w_0 = 0$.
- A non-zero w_0 moves it parallel to itself, away from the origin. In Figure 1 (right), $2x_1 + 3x_2 = 0$ passes through the origin, while $2x_1 + 3x_2 - 6 = 0$ crosses the x_2 axis at $-(-6)/3 = 2$.

So we often take the simplifying assumption that the hyperplane passes through the origin. The equation then shrinks to

$$w^\top x = 0$$

In 2D that is $w_1x_1 + w_2x_2 = 0$; in 4D, $w_1x_1 + w_2x_2 + w_3x_3 + w_4x_4 = 0$.

Extra: The assumption costs nothing, because of a standard trick. Add a constant component $x_0 = 1$ to every point, so $x = [1, x_1, \dots, x_n]$, and put w_0 into $w = [w_0, w_1, \dots, w_n]$. Then $w^\top x = w_0 + w_1x_1 + \dots + w_nx_n$, and the equation $w^\top x = 0$ already contains the shift. This is the column of ones in the matrix form of the [multiple linear regression maths Note](#).

6. What w means: the normal vector

Key point: $w^\top x = 0$ says w makes a 90° angle with every point x on the hyperplane, so w is perpendicular to the hyperplane.

6.1 The argument

Key point: A dot product of non-zero vectors is 0 only at 90° .

For a hyperplane through the origin, every point x on it satisfies $w^\top x = 0$. Write the dot product in its geometric form:

$$w^\top x = w \cdot x = \|w\| \|x\| \cos \theta = 0$$

For non-zero w and x , the lengths are positive, so $\cos \theta = 0$ and $\theta = 90^\circ$. Each point x on the hyperplane is a vector lying in the hyperplane, and w is at a right angle to every one of them. So w is perpendicular to the hyperplane itself. A vector perpendicular to a line, plane or hyperplane is called its **normal vector**.

With numbers, for the line $2x_1 + 3x_2 = 0$ in Figure 1 (left): $w = [2, 3]$, and the points $[3, -2]$ and $[-1.5, 1]$ lie on the line. Both give $w \cdot x = 0$:

$$2 \times 3 + 3 \times (-2) = 0, \quad 2 \times (-1.5) + 3 \times 1 = 0$$

6.2 The same in 3D

Key point: The coefficients of a plane's equation, read as a vector, stick straight out of the plane.

The plane $x_1 + 2x_2 + 2x_3 = 0$ and its normal vector w

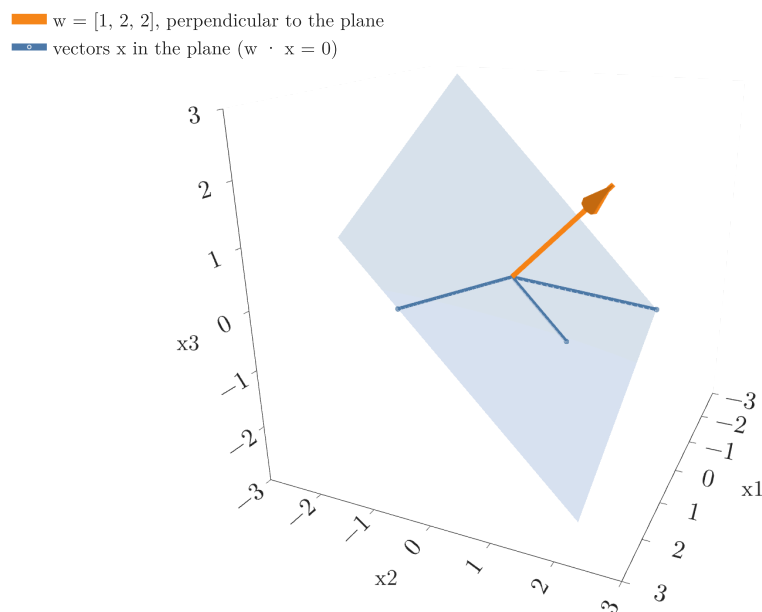


Figure 2: The plane $x_1 + 2x_2 + 2x_3 = 0$ and its normal vector $w = [1, 2, 2]$

Figure 2 shows the plane $x_1 + 2x_2 + 2x_3 = 0$. Its coefficients form $w = [1, 2, 2]$, drawn in orange. The three blue vectors lie in the plane, for example $[2, -1, 0]$, with $1 \times 2 + 2 \times (-1) + 2 \times 0 = 0$. The orange w stands at 90° to all of them.

So reading a hyperplane's equation tells us its direction at once: the coefficients are the normal vector. In nD we cannot draw it, but the argument of Section 6.1 never used the number of dimensions.

Python: Checking that points lie on a hyperplane is one dot product per point.

```
import numpy as np
```

```
w = np.array([1, 2, 2])
```

```
X = np.array([[2, -1, 0], [0, 1, -1], [-2, 2, -1]])
```

```
# dot product of every row of X with w
```

```
X @ w      # array([0, 0, 0]): all on the plane
```

Extra: w stays perpendicular when $w_0 \neq 0$. Take any two points x and y on the hyperplane: $w^\top x + w_0 = 0$ and $w^\top y + w_0 = 0$. Subtracting, $w^\top(x - y) = 0$. The vector $x - y$ runs along the hyperplane, so w is perpendicular to every direction in it. This is why the two lines in Figure 1 (right) share the same w and are parallel.

Extra: For a point off the hyperplane, $w^\top x + w_0$ is not zero, and its sign says which side the point is on: positive on the side w points to, negative on the other. To see this, start at a point p on the hyperplane and step a distance t along w : $x = p + t w / \|w\|$. Then $w^\top x + w_0 = (w^\top p + w_0) + t w^\top w / \|w\| = 0 + t \|w\|$, which has the sign of t . This is the side test of the [perceptron trick Note](#) and the decision rule of the [SVM maths Note](#).

7. Summary

Object	Equation	Normal vector
Line (2D)	$w_1x_1 + w_2x_2 + w_0 = 0$	$[w_1, w_2]$
Plane (3D)	$w_1x_1 + w_2x_2 + w_3x_3 + w_0 = 0$	$[w_1, w_2, w_3]$
Hyperplane (n D)	$w^\top x + w_0 = 0$	w
Through the origin	$w^\top x = 0$	w

- The general form of a line grows into a hyperplane by adding one term per dimension; the sum of terms is a dot product.
- w_0 shifts the hyperplane; $w_0 = 0$ means it passes through the origin.
- w is the normal vector: perpendicular to the hyperplane, because $w \cdot x = 0$ means a 90° angle.

8. Key terms

Term	Meaning
Equation of a hyperplane	$w^\top x + w_0 = 0$: one equation for a line, plane or hyperplane in any dimension
w_0	The constant term; it shifts the hyperplane away from the origin, and is 0 when the hyperplane passes through the origin
Normal vector	A vector perpendicular to a line, plane or hyperplane; for $w^\top x + w_0 = 0$ it is w